BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

# LESSON 213 · Timed Paper 1 data and logic practice

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Which sort builds an ordered prefix by inserting a key?

Q2. How do you compensate for shifting a positive fractional mantissa left?

Q3. Simplify ¬(A ∨ B).

Q4. Find the decimal value of $0.1100_2 \times 2^3$.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

sign extension — Copy the sign bit into a new high-order position.

De Morgan — Swap AND/OR while negating components.

normalisation — Remove redundant leading mantissa sign bits.

mantissa — The fractional significant part.

exponent — Power-of-two scale.

## New learning

Show positive first — Use exactly eight bits for +11, rather than padding after converting.

Invert each bit — 00001011 becomes 11110100.

Add one — 11110100 + 1 = 11110101 for −11.

## Worked example

Model −6 in eight-bit two's complement.

1. Predict first — Write +6 as 00000110.

2. Check the method — Invert to 11111001.

3. Check the method — Add one to get 11111010, representing −6.

# We Do · Practise with guidance

Find −9 in eight-bit two's complement.

`+9=00001001; invert; add 1.`

Predict first

Write 9 as 00001001.

Check the method

Invert to 11110110.

Add one to get 11110111, representing −9.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

(a) Represent −19 as an 8-bit two's-complement number. [4] (b) Perform an arithmetic right shift by 1 of your result. State its signed decimal interpretation. [4]

### Question 2

(a) Use De Morgan's law to simplify ¬(A ∧ ¬B). [3] (b) Explain how an eight-cell group of ones simplifies a three-variable Karnaugh map. [5]

### Question 3

A positive two's-complement fractional mantissa is $0.0011_2$ with exponent 5. Normalise it, explaining each shift and why the value is unchanged.

## Exit Ticket · Answer before checking

An arithmetic right shift of −19 by one place produces −9.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Insertion sort.

Q2. Decrease the exponent by the same number of places.

Q3. ¬A ∧ ¬B.

Q4. 0.75×8=6.

### You Do pages · Independent questions

Q1. [8] (a) +19=00010011 [1], invert=11101100 [1], add 1=11101101 [2]. (b) Arithmetic right shift replicates leading 1 [1], gives 11110110 [2], interpreted as −10 [1]; arithmetic right shift of odd negative values rounds towards negative infinity.

Q2. [8] (a) ¬A ∨ ¬¬B [2], simplifying to ¬A ∨ B [1]. (b) The group covers all eight 3-variable combinations [2]; all three variables vary [1], leaving no literal [1], so output is constant 1 [1].

Q3. [8] $0.0011\times2^{5} \rightarrow 0.0110\times2^{4}$ [2] $\rightarrow 0.1100\times2^{3}$ [2]; positive normalised mantissa begins 01 [1]; shifting left doubles mantissa and reducing exponent compensates [2]; decimal value 6 [1].

### Exit Ticket · Model answer

False. It produces −10 in two's complement due to sign extension and rounding towards negative infinity.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding